Hardware-aware on-chip learning in an analog spiking neural network.*

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Abstract

Hardware-aware on-chip learning in an analog spiking neural network using measured memristive synapses and neurons, with a compact device model and a small fabricated demonstrator.

Programme: UKRI AI Centre for Doctoral Training in AI for Sustainability, University of Southampton

1 Document purpose

This report covers the summer project undertaken. It covers the core elements specified in the brief: introduction, literature review, proposed methodology, impact assessment, plan of future work, references, and a timeline.

1.1 Suggested word budget

Section	Suggested words	Notes
Introduction	400–700	Topic framing, motivation, scope.
Literature Review	1,500–2,000	Critical review, gap identification, research objectives.
Proposed Methodology	800–1,200	Methods, data, tools, feasibility, training needs.
Impact Assessment	600–900	RRI, sustainability, SDGs, ethics, stakeholders.

*Supervised by Dr. Firman Simanjuntak and Prof. Mark Zwolinski, University of Southampton.

Section	Suggested words	Notes
Plan of Future Work	700–1,000	3-year PhD plan and first 9 months in detail.
References	Variable	Not usually counted in the main word limit unless instructed otherwise.

2 Introduction

2.1 Research context

Neuromorphic computing seeks to emulate the brain's event-driven, massively parallel processing using energy-efficient hardware models such as spiking neural networks (SNNs). Memristor device architectures are promising implementations with bio-plausible behaviour, because drift memristors naturally implement analog synaptic weights and non-volatile (long-term) plasticity, while diffusive memristors provides short term neuron firing behaviour.

2.2 Problem statement

The Internet of Things (IOT) requires intelligent information processing in the field, away from powerful and energy-consuming cloud and datacentre-based AI capabilities. Without low-energy local AI inference, very large amounts of data need to be transitted for central processing, requiring energy, reliable bandwidth and connectivity, latency, and the provision of land, energy grid and water supplies.

IOT devices are often disconnected from energy grids and consequently have a limited energy budget, constrained by energy harvesting and battery capacity. A lack of energy self-sufficiency requires ongoing human involvement in battery maintenance and replacement.

2.3 Motivation and significance

Although the field of machine learning (ML) is considered mature, most of the work achieved so far assumes ever-larger and more powerful models running on centralised hardware, with the primary aim of improving accuracy against well-established benchmark datasets and problems. Power demands of frontier models (especially for training) are rising exponentially and will become the limiting factor for further adoption; datacentres cannot be built because of the lack of energy grid and cooling water availability.

The human brain is often quoted for its extraordinary efficiency, with consumption of 20w compared with the kilo- or mega-watts of power needed to operate a traditional computer of equivalent capability. The brain is the inspiration for the recent Neuromorphic field of computing, which aims for in-memory computing (IMC) to circumvent the movement of data from memory to CPU in the traditional Von-Neumann architecture, and thereby remove the associated power consumption.

Recent work has demonstrated memristor CMOS neuromorphic devices with spike-timing-dependent-plasticity (STDP) learning at the synapse level, and small-scale SNNs with on-chip adaptation, but most demonstrations either remain shallow, rely on ex-situ weight programming, or lack tight integration with digital neuromorphic platforms such as FPGAs.

Power consumption per inference in extremely constrained environments is measured, but often only for single devices (synapses, neurons), not for complete systems.

This project aims to bridge these developments by designing, fabricating, and evaluating a multilayer memristor-based SNN with genuine on-chip learning. The focus will be on event-based vision or time-series tasks relevant to ultra-low-power, DVS-free neuromorphic hardware.

2.4 Scope of the summer transition project

The goal of the summer project is to identify a reasonable scope for the PhD research. Many of the concepts are new to me, and I need to gain an understanding of practical laboratory techniques (device probing, measurement, characterisation, design and nano-fabrication).

2.5 Chapter overview

This report covers the summer project undertaken. It covers the core elements required: introduction, literature review, proposed methodology, impact assessment, plan of future work, references, and a timeline.

3 Literature Review

3.1 Search strategy

[Document how the literature search was performed: databases, keywords, time windows, inclusion/exclusion criteria, and any review logic used.] A literature search was performed using IEEE Xplore and Arxiv.

3.1.1 Core search dimensions

Topic focus: - “spiking neural network” OR “SNN” - “neuromorphic computing” OR “neuromorphic hardware” - “on-chip learning” OR “in-situ learning” OR "hardware learning" - "memristor" OR "memristive synapse" OR "crossbar array" - "ferroelectric" OR "antiferroelectric" devices in SNNs

Learning rules and algorithms: - "STDP" OR "spike-timing-dependent plasticity" - "three-factor learning" OR "eligibility trace" OR "e-prop" - "Hebbian learning" OR "Oja's rule" in SNNs - "surrogate gradient" OR "direct feedback alignment" for SNN training

Neuron and synapse models: - "leaky integrate-and-fire" OR "LIF" - "Izhikevich neuron model" OR "Hodgkin–Huxley" (for biological grounding) - "conductance-based synapse" OR "current-based synapse" - "short-term plasticity (STP)" OR "long-term plasticity (LTP)"

Hardware implementation: - "analog neuromorphic" OR "mixed-signal VLSI" - "ASIC" OR "FPGA" OR "neuromorphic processor" - "in-memory computing" OR "compute-in-memory (CIM)" - "event-driven" OR "asynchronous" architectures

Applications and benchmarks: - "event-based vision" OR "DVS camera" - "spike-based classification" OR "temporal coding" - "edge AI" OR "low-power inference"

3.2 Background and foundational concepts

3.3 Memristor–CMOS neuromorphic architectures

Hybrid memristor–CMOS systems use memristive devices (e.g. ReRAM cells) to store synaptic weights in crossbar arrays and CMOS neuron circuits to implement spiking dynamics.

Published memristor–CMOS neuromorphic chips have demonstrated on-chip STDP learning in synapses, basic pattern recognition, and small-scale neural cores, often in 130 nm CMOS with sub-micrometre memristor devices.

Reviews of CMOS-compatible memristors emphasise analog tunability, endurance, and integration challenges, and point to neuromorphic computing as a key application area.

3.4 On-chip learning in memristive neural networks

Work on memristor-based neural networks has explored both ex-situ training (ANN/SNN training in software followed by weight programming into crossbars) and in-situ training using local learning rules.

On-chip training schemes for SNNs include STDP variants and supervised approximations to backprop adapted to hardware, where timing and pulse-based updates are critical to realising learning in physical synapses.

Recent studies argue that robust on-chip learning in memristor-based networks requires careful handling of device variability, limited conductance precision, and non-linear update dynamics.

3.5 FPGA-based neuromorphic and SNN systems

FPGAs have been widely used to implement SNNs because they offer reconfigurable logic, DSP blocks, and custom spike routing, enabling real-time neuromorphic processing at the edge.

FPGA-based neuromorphic vision accelerators demonstrate event-driven processing pipelines where SNNs classify or track objects using spike trains derived from image sensors.

These systems provide an excellent platform for rapid architectural exploration, precise timing control, and integration with conventional cameras and other sensors, but they do not typically exploit analog synaptic computation.

3.6 Gaps and originality

There is a clear gap in scalable multilayer memristor SNNs with on-chip learning, particularly where analog synapses are tightly co-designed with a digital neuromorphic platform that can both emulate and train them.

Existing memristor implementations tend to focus on single-layer cores or small networks and rarely provide systematic comparisons against FPGA-based SNN implementations under identical tasks and metrics.

Conversely, FPGA SNN accelerators rarely interface to trainable analog cores, meaning that the potential energy and latency advantages of memristive crossbars are not exploited in end-to-end systems.

This project will address these gaps by designing a multilayer memristor–CMOS SNN core with on-chip learning and integrating it with an FPGA platform for experimentation, training, and system-level evaluation on event-based tasks.

3.7 Current state of the art

[Review the most relevant studies, methods, systems, datasets, benchmarks, policies, or theoretical approaches. Organise by theme rather than paper-by-paper where possible.]

3.8 Overview

This literature review surveys recent work at the intersection of analog spiking neural networks (SNNs), memristive and ferroelectric/antiferroelectric devices, and hardware-aware on-chip learning, with a focus on compact device models and fabricated demonstrators. It organizes the field into five main themes: 1. Device-level advances in memristive and ferroelectric synapses and neurons. 2. Analog and mixed-signal neuromorphic circuits for SNNs. 3. On-chip and hardware-aware learning rules. 4. System-level neuromorphic demonstrators and reservoir/edge systems. 5. Surveys and reviews that contextualize these developments.

3.9 Device-Level Memristive and Ferroelectric Synapses

- Memristor synapses provide non-volatile or semi-volatile conductance states with analog or multi-level programmability, enabling in-memory computing and synaptic plasticity in SNNs.[1]
- Recent device-level reviews emphasise short-term and long-term plasticity, spike timing-dependent plasticity (STDP), spike rate-dependent plasticity, and even optoelectronic control as critical behaviours for neuromorphic synapses.[1]
- Chandrasekaran et al. present a device-level critical review of memristor synapses, combining a statistical survey of neuromorphic device publications (2018–2025) with detailed discussion of biomimetic properties such as STP, LTP, STDP, and SRDP in various material systems.[1]
- They highlight optical-stimulus-driven 2D-material optoelectronic synapses and application examples ranging from MNIST recognition to hardware pattern recognition and biomedical interfaces, underlining the diversity of physical mechanisms that can underpin synaptic plasticity.[1]
- Jin et al. provide a broader review of memristor-based artificial neural networks for hardware neuromorphic computing, covering convolutional, recurrent, and spiking networks, and emphasising memristor in-memory computing as a way to overcome the von Neumann bottleneck.[1]
- Their discussion of synaptic plasticity mechanisms (potentiation, depression, and emerging learning algorithms tailored to memristive dynamics) is directly relevant to hardware-aware learning rules in SNNs, especially when device non-idealities constrain achievable weight trajectories.[1]

3.10 Ferroelectric and Antiferroelectric Synaptic Arrays

- Ferroelectric memristor crossbar arrays offer high-density, energy-efficient synaptic matrices with tunable tunnelling electroresistance and switching dynamics, enabling both short-term and long-term memory effects.[1]
- Park et al. report a 24×24 ferroelectric memristor crossbar array based on a TiN/HAO/SiO2/n-Si structure that demonstrates scalable device characteristics, improved TER ratios in smaller cells, and neuromorphic functionalities such as paired-pulse facilitation and spike-duration/number-dependent plasticity.[1]
- The same work uses the ferroelectric array as a reservoir layer in a reservoir computing system, achieving high accuracy on MNIST and Fashion-MNIST, and demonstrating associative learning and nociceptor-like behaviour, thereby validating the array as a multifunctional synaptic substrate.[1]
- The reported device-to-device and cycle-to-cycle uniformity addresses a central challenge in memristive arrays, making these ferroelectric synapses

particularly attractive for hardware-aware learning where variability and drift must be explicitly modelled.[1]

- Lu et al. review ferroelectric-based leaky integrate-and-fire (LIF) neurons and outline device design and performance optimisation strategies, emphasising the realization of leak via depolarization or antiferroelectric phases.[1]
- Their identification of key metrics—hardware cost, energy consumption, endurance—and application examples provides a complementary perspective to ferroelectric synapses, highlighting co-design opportunities between devices implementing integrate/leak and those implementing synaptic weight storage.[1]

3.11 Memristive and Ferroelectric Neurons

- A series of recent works demonstrate compact artificial neurons implemented directly in memristive or ferroelectric devices, avoiding large external capacitors and complex reset circuitry while achieving rich spiking dynamics.[1]
- Zhang et al. realise a compact artificial memristive neuron with leaky integration, automatic threshold firing, and self-recovery in a single device with ultralow energy consumption, providing a device-level LIF neuron building block.[1]
- Cao et al. introduce a compact neuron based on an antiferroelectric field-effect transistor (AFeFET) using Hf0.2Zr0.8O2, where intrinsic polarisation and spontaneous depolarisation implement integrate-and-leak behaviour, achieving low spike energy, high endurance, and strong uniformity.[1]
- They further construct a two-layer fully ferroelectric SNN combining AFeFET neurons with ferroelectric synapses, reaching high MNIST recognition accuracy and providing a fully ferroelectric neuromorphic demonstrator.[1]
- Gao et al. extend antiferroelectric transistor concepts to reconfigurable neuromorphic functions via coupled polarisation switching and charge trapping, enabling multiple neuromorphic behaviours (e.g., synaptic and neuronal states) in the same device.[1]
- Shi et al. present coplanar floating-gate antiferroelectric transistors that unify volatile neuron-like responses, nonvolatile synaptic behaviour, and fading-memory reservoir dynamics in a single device architecture, culminating in an all-in-one analog reservoir computing system.[1]
- Zhidkov et al. demonstrate multiple spiking modalities (time-to-first-spike, spike count, firing-rate coding) in annealing-optimised Ag/Hf0.5Zr0.5O2 memristive neurons, realised using filamentary switching and a simple series resistor, illustrating how process engineering can tune spiking behaviour.[1]

- Wang et al. focus on threshold-switching memristors and review their use as spiking neurons with oscillatory, LIF, Hodgkin–Huxley-like, and stochastic dynamics, directly linking device physics (Mott vs. redox switching) to neuron models and energy consumption.[1]
- Zhao et al. present a spiking artificial neuron based on one diffusive memristor, one transistor, and one resistor, realising compact neuron circuits with nanoscale footprint and integrating them within larger neuromorphic systems.[1]
- Kim et al. show self-powered analog neuromorphic systems for multimodal sensing and learning using diffusive and drift memristors, where neuromorphic sensing, encoding, and learning are all co-located in hardware, underscoring opportunities for fully integrated edge systems.[1]

3.12 Memristive Crossbars and Mixed-Signal Perceptrons

- Passive memristive crossbars (0T1R) and active 1T1R arrays underpin many in-memory computing architectures for ANNs and SNNs, providing high-density weighted-sum operations and local update paths.[1]
- Bayat et al. demonstrate a one-hidden-layer perceptron classifier implemented entirely in mixed-signal hardware using two passive 20×20 metal-oxide memristive crossbars, with ex-situ training and classification fidelity close to simulations despite device non-idealities.[1]
- Poddar et al. report ultra-high density perovskite nanowire array memristors for multi-layer perceptron networks, emphasising nanowire crossbars as a path to extremely compact synaptic fabrics.[1]
- Jin et al. and Mehonic et al. discuss memristor-based multilayer networks more generally, including full-memristive neural networks where both neurons and synapses are implemented in memristors, exploring algorithmic adaptations of delta rule and backpropagation to such hardware.[1]
- A recent hybrid convolutional SNN architecture by Go and Kim combines digital non-leaky integrate-and-fire neurons with Knowm self-directed channel memristor-based synapses in a 1T1R crossbar, using a current-sense amplifier and 1-bit comparator as a minimal readout instead of high-resolution ADCs.[1]
- Their intensity-to-latency temporal coding scheme reduces spike counts and device stress, and they analytically compare energy usage of this minimalist readout versus conventional ADC-based designs, which is central to hardware-aware learning in edge contexts.[1]

3.13 Analog and Mixed-Signal Neuron Circuits

- On the CMOS circuit side, Wu et al. present a reconfigurable leaky integrate-and-fire neuron with a single op-amp that accommodates a large number of memristor synapses and enables online STDP learning with optimised power consumption.[1]
- Their 180 nm CMOS design achieves high power-efficiency for STDP in large synapse populations and low integration current, making it a general-purpose building block for dense memristor-synapse neuromorphic systems.[1]
- Earlier silicon neuron work such as the classical "silicon neuron" and compact neuron supplements in the bibliography provide historical context and baseline CMOS neuron architectures that current memristive and ferroelectric neuron designs build upon.[1]
- These CMOS designs typically rely on capacitors and current-mode implementations, motivating the recent shift towards device-intrinsic capacitance and polarisation dynamics in ferroelectric neurons to reduce area and static energy.[1]

3.14 Learning Rules and On-Chip Training in SNNs

- Biological and bio-inspired plasticity rules such as STDP, Hebbian learning, and voltage-dependent synaptic plasticity are central to on-chip learning in SNNs, especially when learning must be implemented locally using device physics.[1]
- Diehl and Cook demonstrate unsupervised digit recognition using STDP, lateral inhibition, and adaptive thresholds without any labels or teacher signals, achieving high MNIST accuracy and showing the robustness of biologically plausible mechanisms across different learning rules.[1]
- Goupy et al. introduce Stabilized Supervised STDP (S2-STDP) combined with Paired Competing Neurons (PCN) for classification layers in SNNs, integrating error-modulated weight updates with local spike timing and intra-class competition.[1]
- They show improved performance over previous supervised STDP rules across multiple vision datasets, and their architecture is particularly amenable to hardware implementation due to its locality and lack of global backpropagation.[1]
- Recent work on Hebbian enrichment of SNNs proposes memory-augmented architectures where Hebbian synaptic plasticity acts as both memory and processor, improving out-of-distribution generalisation, one-shot learning, and reward-based learning.[1]

- These Hebbian-enriched SNNs underline the computational versatility that can be achieved when synaptic updates are tightly coupled to local spike statistics, which is attractive for analog on-chip learning with memristive devices.[1]
- At the hardware level, energy- and endurance-aware co-design for neuromorphic edge intelligence highlights the need to tailor learning algorithms to device constraints such as limited endurance, stochastic switching, and restricted dynamic ranges.[1]
- Hybrid schemes that split feature extraction (often unsupervised STDP) from supervised classification layers and minimise write operations to memristors exemplify hardware-aware learning strategies that your proposed work aims to extend in a fully analog context.[1]

3.15 Threshold-Switching Memristors and Spiking Dynamics

- Threshold-switching memristors (TSMs) with volatile or semi-volatile dynamics naturally implement neuron-like behaviours such as oscillations, LIF spiking, and refractory periods through their intrinsic switching physics.[1]
- Wang et al. review TSM-based spiking neuromorphic systems, cataloguing device materials (Mott insulators, valence-change memories, diffusive systems) and associated spiking modes, along with energy-per-spike figures and neuromorphic platform demonstrations.[1]
- Li et al. present Bi2Se3-based threshold-switching memristors with high switching ratios, low threshold voltages, and thermoelectric response, using HSPICE simulations to explore spiking oscillation responses to temperature variations and resistance states.[1]
- This work targets neuromorphic thermoreception, illustrating how TSM neurons can be integrated with sensory modalities in hardware, and how compact device models capture combined electrical and thermal dynamics relevant to edge sensing.[1]

3.16 Neuromorphic Reservoirs and All-in-One Architectures

- Reservoir computing offers a flexible framework for temporal information processing using fixed recurrent substrates and trained readouts, which maps well onto analog device arrays with rich internal dynamics.[1]
- Ferroelectric memristor arrays and coplanar FG AFeFETs have been used as physical reservoirs, exploiting volatile and fading-memory behaviours to encode temporal input patterns, with external or on-chip readouts performing linear classification.[1]

- Shi et al.'s coplanar FG AFeFET architecture exemplifies an all-in-one analog reservoir computing system, where devices can be configured as neurons, synapses, or reservoir nodes depending on biasing and layout.[1]
- The reported recognition accuracies on MNIST and Fashion-MNIST, coupled with tunable volatility and non-volatility, show that single-device reservoirs can achieve competitive performance while remaining area-efficient and compatible with CMOS back-end integration.[1]

3.17 Neuromorphic Edge and Embedded Applications

- Several works in the bibliography focus on SNNs as controllers or embedded intelligence for resource-constrained platforms, such as FPGA-based neuromorphic systems and spiking neurocontrollers for UAVs.[1]
- Qiu et al. evolve spiking neurocontrollers for full hexacopter control in six degrees of freedom, using modular decomposition and NEAT-style evolution, highlighting how spiking networks can be tailored to continuous control tasks.[1]
- Javanshir et al. review advances in algorithms and neuromorphic hardware for SNNs, including FPGA, digital, analog, and hybrid platforms, and survey SNN simulators and training methods.[1]
- At the edge-intelligence level, Go and Kim's hybrid CSNN and Kim et al.'s self-powered multimodal system underline the importance of co-designing device physics, circuit architecture, and coding schemes to minimise energy while maintaining accuracy.[1]

3.18 Surveys and Contextual Reviews

- Schuman et al.'s survey of neuromorphic computing provides a long-range historical and thematic overview spanning neuro-inspired models, learning approaches, hardware/devices, supporting systems, and applications, clarifying how memristive and ferroelectric neuromorphic hardware fits into the broader landscape.[1]
- A subsequent article on opportunities for neuromorphic computing algorithms and applications by the same group highlights open challenges in algorithm–hardware co-design, robustness, and application-driven benchmarks.[1]
- Mehonic et al. discuss memristors from in-memory computing and deep learning acceleration to SNNs and future neuromorphic systems, emphasising the need for tailored learning and inference algorithms that exploit non-von-Neumann architectures.[1]
- Javanshir et al. and other reviews on SNN algorithms, neuromorphic hardware platforms, and FPGA implementations complement device-centric

surveys by providing algorithmic and architectural perspectives.[1]

3.19 Synthesis and Gaps Relative to the Target Topic

- The attached bibliography shows strong activity in device-level memristive and ferroelectric neurons and synapses, ferroelectric LIF neuron design, threshold-switching memristor-based spiking dynamics, and mixed-signal memristive crossbar demonstrators.[1]
- It also includes several hardware-aware learning schemes—unsupervised STDP, supervised STDP variants, Hebbian-enriched networks, and endurance-aware co-design—but these are typically studied either in simulation or with limited on-chip learning in small arrays.[1]
- Compact device models are commonly used in circuit-level simulations (e.g., HSPICE models for Bi2Se3 TSMs, compact neuron models in memristive and ferroelectric devices), yet there remains a gap in unified compact models that simultaneously capture neuron dynamics, synaptic plasticity, and non-idealities for multi-layer analog SNNs.[1]
- Fabricated demonstrators often focus either on device-level proof-of-concept (single neurons, small crossbars, reservoir layers) or on mixed-signal systems with ex-situ training; fully analog, multi-layer SNNs with hardware-aware on-chip learning remain comparatively underexplored.[1]
- Your proposed subject—"hardware-aware on-chip learning in an analog SNN using measured memristive synapses and neurons, with a compact device model and a small fabricated demonstrator"—naturally builds on these strands by combining measured device characteristics, compact unified models, and on-chip learning rules tuned to device constraints.[1]
- The literature indicates that key open problems include handling device variability and stochasticity, designing plasticity rules compatible with endurance limits, integrating ferroelectric and memristive devices in multi-layer analog SNNs, and demonstrating small yet complete hardware prototypes that close the loop between learning, device physics, and circuit behaviour.[1]

3.20 Critical analysis of existing work

[Evaluate strengths, limitations, assumptions, methodological weaknesses, contradictory findings, and open issues.]

3.21 Identified research gap(s)

[State clearly what is missing in the literature. Be explicit about the unresolved technical, scientific, methodological, or application-level gaps.]

3.22 Research objectives and questions

3.22.1 Success criteria

The project will successful if it demonstrates all three: - Physical credibility: Learning-rule simulations use empirical distributions from physical devices, not ideal device behaviour. - Mechanism insight: Why a device feature helps or harms learning, and how circuit or algorithm design mitigates it. - Practical evidence: at least one experiment updates physical states in response to spike-derived signals and improves a measurable task objective.

3.22.2 RQ1 – CMOS-compatible scalable SNN architecture

What memristor–CMOS co-designs for synapses and neurons enable a scalable, multilayer spiking neural network that can be fabricated using the CMOS processes available at Zepler? Within this constraint, which crossbar topologies, neuron circuit models (e.g. LIF variants), and spike-routing schemes jointly optimise trainability, energy per spike, and silicon area for edge-scale event-based workloads?

3.22.3 RQ2 – Efficient on-chip learning for multilayer memristor SNNs

Which on-chip learning algorithms—such as STDP/VDSP and hardware-feasible approximations to spiking backpropagation—can be implemented efficiently in multilayer memristor SNNs, and how do their accuracy, convergence speed, and energy per update compare to ex-situ training with weight transfer under realistic device non-idealities?

3.22.4 RQ3 – Robustness and exploitation of device variability

What levels of device-to-device and cycle-to-cycle variability in memristor synapses and neurons can a multilayer SNN tolerate before task accuracy degrades beyond a defined threshold, and how can on-chip learning rules be designed so that such variability either remains harmless or actively improves training speed and final accuracy?

3.22.5 RQ4 – System-level performance on event-based tasks

Can a multilayer memristor SNN with on-chip learning achieve competitive task accuracy, energy per event, and latency on representative event-based vision or time-series benchmarks compared with state-of-the-art software ANN/SNN baselines, when implemented under Zepler-compatible CMOS and realistic memristor non-idealities?

3.23 Novelty and significance

[Explain what is novel about the proposed direction and why it would matter if successful.]

4 Proposed Methodology

4.1 Methodological overview

[Describe the overall research design and justify why it is suitable for addressing the identified gaps.]

4.2 Proposed methods

[Detail the methods, models, experiments, analytical techniques, simulations, fieldwork, datasets, case studies, or design processes you expect to use.]

4.2.1 Diffusive memristor characterisation

Use existing fabricated diffusive memristors to characterise device behaviour, with a particular focus on device-to-device, cycle-to-cycle and stochastic variability ("variability statistical model").

Implement variability model in software emulation of a single, simple neuron. Convert memristor variability into neuron behaviour variability.

Implement simulation of on-chip learning (STDP and VDSP) in software, for single, simple neurons (pre/post-synaptic), initially without variability. Introduce the variability statistical model created earlier.

Evaluate the impact of variability on learning speed and accuracy.

Move to a larger network, where feature extraction and differentiation becomes important. Understand whether variability assists in feature / node differentiation is assisted/accelerated by variability, or whether there is a point at which excessive variability becomes catastrophic for network performance.

Define an allowable variability specification.

4.2.2 Memdevice and circuit design

Use existing memdevice expertise to select, design and fabricate CMOS-compatible memristor stacks with suitable characteristics (tuning, endurance, variability, and process compatibility).

Design hybrid synapse cells (e.g. 1S1R) and neuron circuits (e.g. LIF 1M1T1R neurons) that operate at realistic voltage ranges and support pulse-based conductance updates implementing STDP/VDSP or supervised learning rules.

Use SPICE/Verilog-A simulation to validate crossbar and neuron behaviour, including parasitic effects and sneak paths.

4.2.3 Compact modelling and multi-scale simulation

Characterise fabricated devices and arrays to obtain IV curves, switching behaviour, endurance, retention, and variability, then fit compact models suitable for both circuit and network-level simulation.

Integrate these models into a multi-scale simulation stack linking device-level dynamics to spiking network behaviour, using Python SNN frameworks for training and analysis.

4.2.4 On-chip learning schemes

Implement unsupervised STDP/VDSP-based learning using local timing and pulse protocols that update memristor conductances based on spike correlations and presynaptic potentials.

Explore supervised learning schemes adapted to hardware, such as approximate backpropagation in the spiking domain, where error signals are encoded as additional spike trains or weight-update pulses orchestrated by the FPGA controller.

Compare ex-situ training (software SNN trained first, then weights programmed into crossbars) to in-situ training (updates performed directly in hardware) under realistic device constraints.

4.2.5 Evaluation and analysis

Evaluate accuracy, energy per event, latency, and robustness on chosen tasks; compare across analog SNN, FPGA SNN, and software baselines using consistent metrics and datasets.

Perform sensitivity analyses to quantify the impact of device variability, limited conductance precision, and learning rule parameters on long-term performance and stability.

4.3 Data, resources, and infrastructure

[Identify expected data sources, software, hardware, facilities, lab access, compute requirements, and any external dependencies.]

4.4 Evaluation strategy

[Explain how success will be measured: metrics, baselines, validation strategy, comparison methods, robustness checks, or qualitative evaluation.]

4.5 Risks, assumptions, and limitations

[Identify key risks and assumptions, then explain mitigation plans and fallback options.]

4.6 Skills audit and training needs

[Identify technical or professional skills gaps. Link these to a training plan for the PhD phase.]

Skill / capability gap	Why it matters	Planned training or action	Indicative timing
Device measument	Needed to characterise devices for software simulation	B53 measurement lab induction, training	Sept 2026
Device fabrication	Needed to fabricate devices for physical evaluation	B53 nano fab lab induction, tool training	Oct-Dec 2026

5 Impact Assessment

5.1 Overview of anticipated impact

[Describe the potential academic, environmental, industrial, societal, and policy relevance of the proposed research.]

5.2 Responsible Research and Innovation

[Assess the project using the RRI ideas taught in SUST6001.]

5.3 4Ps framework

[Discuss the relevant 4Ps dimensions as taught in your programme, and explain how they apply to this project.]

5.4 AREA framework

[Structure the discussion under Anticipate, Reflect, Engage, and Act/Respond, as appropriate to your course framing.]

AREA element	Project-specific considerations	Proposed actions
Anticipate	[Potential impacts, risks, uncertainties]	[Actions]
Reflect	[Assumptions, values, blind spots]	[Actions]
Engage	[Stakeholders to involve]	[Actions]
Act / Respond	[How project design may adapt]	[Actions]

5.5 Sustainability considerations

[Explain how sustainability is embedded in the project, including lifecycle, energy, materials, deployment, accessibility, social sustainability, or systems-level effects where relevant.]

5.6 Relevant Sustainable Development Goals

[Identify and justify the SDGs most relevant to the project.]

SDG	Relevance to project	Evidence / rationale
[SDG number and title]	[Why relevant]	[Short justification]

5.7 Ethics and governance

[Note any ethical issues, regulatory requirements, governance concerns, data protection issues, safety implications, or approvals that may be needed.]

5.8 Stakeholders and beneficiaries

[Identify who may benefit, who may be affected, and how stakeholder perspectives may shape the research.]

6 Plan of Future Work

6.1 Overview of PhD direction

[Summarise the intended overall trajectory of the PhD over the 3-year research phase.]

6.2 Detailed plan for the first 9 months

[Provide a detailed plan leading up to the first progression report, including literature work, technical setup, pilot studies, methods development, supervisory engagement, and early outputs.]

Month	Planned activities	Deliverables / milestones
1	Complete formal literature review on memristor neuromorphic devices, on-chip learning in SNNs	Review
2	Measurement training]	[Outputs]
3	Nano fab training	[Outputs]
3	Poster at Neumat London November 2026	Poster
4	Characterise synapse and neuron devices	[Outputs]
5	Build software simulation of 2 synapses / 1 neuron, with device variations	[Outputs]
6	Evaluate on-chip learning methods, showing device variation impact	[Outputs]
7	Extend simulation to multilayer network	[Outputs]
8		[Outputs]
9	[Activities]	[Outputs]

6.3 Milestones for years 1 to 3

[Set out major milestones across the full PhD period, including formal progression points and research outputs.]

6.4 Year 1

Literature review.

Device measurement and fabrication training completion.

Device characterisation of drift memristors for synapses and diffusion memristors for neurons.

Design and implementation of simple (single, 2 synapse/ 1 neuron) and multilayer SNN simulations; establishment of software evaluation pipeline.

Architectural specification for the memristor SNN core and initial simulation models.

6.5 Year 2

Design and simulation of memristor synapse and CMOS neuron circuits; layout of small crossbar arrays.

Fabrication of test structures at Zepler; device and array-level characterization; development of compact models.

6.6 Year 3

Implementation of on-chip learning schemes (STDP and supervised variants) in the memristor core; demonstration of learning on small tasks.

Integration of FPGA platform and analog SNN core; end-to-end demonstrations with real inputs; comparative evaluation against FPGA-only and software SNN baselines.

Sensitivity and robustness studies; derivation of design guidelines and preparation of main publications.

Consolidation of results into the thesis; further publications; optional exploratory work on AFeFET-based synapse/neuron concepts as future extensions.

Period	Milestones	Notes
Year 1	First progression review, pilot results, initial publication target	[Notes]
Year 2	[Confirmation review and major study completion]	[Notes]
Year 3	[Third progression review, thesis writing, publications, completion planning]	[Notes]

6.7 Gantt chart notes

[Insert or link the Gantt chart here, or explain where it is provided. The chart should cover the 3-year PhD period, include detailed planning for the first 9 months, and show key milestones such as progression reviews, publications, and major research stages.][1]

6.8 Meeting and supervision plan

Weekly face-to-face meeting with Dr. Firman Simanjuntak and Prof. Mark Zwolinski

6.9 Contingency planning

[Describe fallback plans if the original research route, data access, technical platform, or experimental setup changes.]

By March 2027, evaluate whether reliable access to a device dataset exists which could anchor a publishable hardware-aware learning result by March 2028. If so, proceed with integrated device-circuit-algorithm development; if not, pivot from fabrication to variability-aware learning rules.

7 References

[Insert references in your required citation style, for example IEEE, APA, Harvard, or a department-approved alternative.]

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8 Appendices

8.1 Appendix A. Summer project timeline

- Start date: w/c 8 June 2026.[1]
- Submission deadline: 11 September 2026.[1]
- Duration: 12 weeks plus 2 weeks of holiday to be agreed with supervisors.[1]
- Possible extensions: typically 1–2 weeks where justified by September exam resits or extenuating circumstances, subject to supervisory approval.[1]

8.2 Appendix B. Submission and assessment notes

- The report is submitted through the SustAI CDT summer project submission form.[1]
- The supervisory team assesses the report first and should provide formative feedback within one month of submission.[1]
- The CDT director checks whether the report and Gantt chart meet minimum requirements.[1]
- The summer project report should later be attached alongside the first progression review submission in the PGR Manager System, where both are considered at the progression review viva.[1]

8.3 Appendix C. Optional paper-style submission variant

[Use this appendix only if the supervisory team agrees that the summer project will be written in paper form rather than thesis-style report form.]

- Proposed paper title.
- Target venue or paper format.
- Abstract.

- Introduction.
- Related work.
- Method / proposed approach.
- Early results or planned evaluation.
- Added impact assessment section.
- Added future work / PhD plan section.

9 Checklist before submission

- ☐ Introduction drafted.
- ☐ Literature review includes a clear gap analysis.
- ☐ Research objectives are explicit and justified.
- ☐ Methodology is feasible and well matched to the research gap.
- ☐ Skills gaps and training needs are identified.
- ☐ RRI discussion includes 4Ps and AREA.
- ☐ Sustainability and SDGs are addressed.
- ☐ First 9 months of the PhD are planned in detail.
- ☐ 3-year Gantt chart is included or attached.
- ☐ Key progression milestones are shown.
- ☐ References are complete and consistently formatted.
- ☐ Supervisor feedback has been incorporated where available.